Advanced Artificial Intelligence: Diffusion Models

CSCI 546

Instructor Info

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Office Hrs: Tues 12:15-1:15, Thurs 12-2

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Course Info

Prerequisites: Ability to code in some language (ideally Python). Integral and Differential Calculus. Previous experience with or strong interest in probability.

Class meetings: Tues, Thurs 10:50-12:05p

Wilson 1123

Course description

This course provides an in-depth study of diffusion-based generative models in machine learning, including score-based models, denoising diffusion probabilistic models (DDPMs), and conditional and spatiotemporal variants. Students will explore the theoretical foundations, including Markov chains, stochastic differential equations, and score matching. Students will also learn how transformers can be used within diffusion modeling. Practical aspects of diffusion modeling will be explored in a final project.

Readings

- *First half of the course*: Here we will focus on mathematics behind diffusion modeling. We will use the textbook:
  Grimmett, G., & Stirzaker, D. (2020). *Probability and random processes.* (4th Edition). Oxford university press.
- *Second half of the course:* Here we will focus on the machine learning of diffusion modeling, and will use the various resources given in the Class Schedule.

Learning Outcomes

- To develop fluency in the mathematics underlying diffusion modeling (probability, Markov chains, and diffusion processes).
- To understand the fundamentals of diffusion modeling in machine learning;
- To apply diffusion modeling to a real or simulated dataset.

Learning Philosophy

This course will use an "inquiry-based learning" approach. This means that classes will not be lecture-oriented. Instead, to quote Dr. Dana Ernst,

> *You will be expected to work actively to construct your own understanding of the topics at hand with the readily available help of me and your classmates. Many of the concepts you learn and problems you work on will be new to you and ask you to stretch your thinking. You will experience frustration and failure before you experience understanding. This is part of the normal learning process. If you are doing things well, you should be confused at different points in the semester.*

Active learning has been shown to increase student performance in STEM courses (e.g. see [1, 2]).

Class activity

A typical class meeting will be structured (approximately) as follows:

Mathematics Module
- Review prev. exercises: 30 mins
- Mini-lecture: 15 mins
- New group exercises: 30 mins

ML Module
- Reading survey: 10 mins
- Peer lightning pres: 10 mins
- Mike's presentation: 25 mins
- Miscellaneous: 25 mins

Daily readings

Before each class meeting, you will be assigned a reading (or task). These tasks are listed on the class schedule at the end of the syllabus. The reading serves as preparation for actively participating in the class that day.

Group exercises

For each class meeting within the mathematics module, students will be split randomly into groups of three to work on problems using the whiteboards. Mathematics is not a spectator sport!

Grading

Course grades will be determined by the decomposition:

- Midterm: 30%
- Group Exercise Participation: 10%
- Reading surveys: 20%
- Presentations: 15%
- Project: 25%

Letter grades will be assigned as follows:
A: 93-100, A-: 90-93, B+: 87-90, B: 83-87, B-: 80-83, C+: 77-80, C: 73-77, C-: 70-73, D+: 67-70, D: 63-67, D-: 60-63, F: 0-60.

Many products (midterm questions, reading surveys, lightning chats) will be graded as follows:

- 4: Clear, correct response demonstrating strong understanding of the concept behind the question.
- 3: Mostly correct with reasonable explanation, but with minor mathematical or conceptual issues.
- 2: Shows partial understanding or some useful intuition, but contains significant errors or gaps in reasoning.
- 1: Minimal understanding demonstrated; response is incomplete, unclear, or only shows limited engagement with the question.
- 0: No meaningful work shown or question left blank.

Reading surveys. The purpose of reading surveys is to check that you did the reading and tried to understand it. Questions will tend towards being open-ended, and will be graded as above.

Makeup policy. It is highly recommended to keep up with the class, as the material is cumulative. However, I drop 2 participation days with no consequence. Missed material beyond that (with a valid explanation and documentation) can be replaced with by increasing the weight of the midterm and final project grades.

Diversity and Inclusivity Statement. I consider this classroom to be a place where you will be treated with respect, and I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Your suggestions about how to improve the value of diversity in this course are encouraged and appreciated. Please let me know ways to improve the effectiveness of the course for you personally or for other students or student groups.

Accommodations for Students with Disabilities. If you are a student with a disability and wish to use your approved accommodations for this course, contact me during my office hours to discuss. Please have your Accommodation Notification available for verification of accommodations. Accommodations are approved through the Office of Disability Services located in 137 Romney Hall. www.montana.edu/disabilityservices.

Student Conduct. You are expected to abide by MSU's Code of Student Conduct.

Scholarly Responsibilities. The class has no homework. However, to be successful in the course, you need to do the following:

- Before class: *Read the assigned section.* Be sure to read actively , with a pencil and paper in hand. For the mathematics module, work out some of the examples within the section on your own, using the textbook to check your work.
- During class: *Attend class.* This allows you to participate in the group activities and discussions.
- After class: *Finish working through the group exercises.* Solutions to group exercises are posted in Canvas. During class time, you will probably just get the gist of some problems. After class, you should make sure you know how to do all of the group exercises.

Class Schedule

MODULE 0: Course Overview

Tues	Jan 13	Course Overview	

MODULE 1: Mathematics

MODULE 1A: Probability Fundamentals

Thurs	Jan 15	Events and their probabilities	GS Sec 1.1-1.7
Tues	Jan 20	Random variables and their distributions	GS Sec 2.1-2.3
Thurs	Jan 22	Random variables and their distributions	GS Sec 2.4-2.6
Tues	Jan 27	Discrete random variables	GS Sec 3.1-3.5
Thurs	Jan 29	Discrete random variables	GS Sec 3.6-3.10
Tues	Feb 3	Continuous random variables	GS Sec 4.1-4.5
Thurs	Feb 5	Continuous random variables	GS Sec 4.6-4.11

MODULE 1B: Markov Chains

Tues	Feb 10	Markov Chains	GS Sec 6.1, 6.2
Thurs	Feb 12	Markov Chains	GS Sec 6.3, 6.4
Tues	Feb 17	Markov Chains	GS Sec 6.5, 6.6, 6.9

MODULE 1C: Diffusion Processes

Thurs	Feb 19	Brownian motion	Chang Sec 5.1-5.6. Suppl: GS Sec 8.1, 8.5
Tues	Feb 24	Diffusion Processes	Chang Sec 6.1-6.5. Suppl: GS Sec 13.3
Thurs	Feb 26	Stochastic Calculus	Chang Sec 6.7-6.8. Suppl: GS Sec 13.7
Tues	Mar 3	Itô integral	Chang Sec 6.9-6.11. Suppl: GS Sec 13.8-13.9

MIDTERM

Thurs	Mar 5	MIDTERM EXAM	

MODULE 2: Machine Learning

MODULE 2A: Introduction to Diffusion

Tues	Mar 10	Variational Inference	Blei, D. M., Kucukelbir, A., & McAuliffe, J. D. (2017). Variational inference: A review for statisticians. *Journal of the American statistical Association*, 112(518), 859-877.
Thurs	Mar 12	Variational Autoencoders	Kingma, D. P., & Welling, M. (2014). Auto-encoding variational Bayes. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
		No Classes — Spring Break	
Tues	Mar 24	Classic Diffusion Model	Sohl-Dickstein, J., Weiss, E., Maheswaranathan, N., & Ganguli, S. (2015, June). Deep unsupervised learning using nonequilibrium thermodynamics. In *International conference on machine learning* (pp. 2256-2265). *Lightning chat*: Josh B. & Racquel
Thurs	Mar 26	Overview of Diffusion Models	Christopher Bishop (2023), *Deep Learning* (Chapter 20: Diffusion Models). Note: for now, only read Secs 20.1 & 20.2. *Lightning chat*: Blake & Matt
Tues	Mar 31	Diffusion exercise	Code up a simple diffusion model! *Reference:* https://e-dorigatti.github.io/math/deep%20learning/2023/06/25/diffusion.html

MODULE 2B: Diffusion and Score Matching

Thurs	Apr 2	Classic Score Matching	Hyvärinen, A. (2005). Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research*, 6(4). Supplemental Exercises: https://ericmjl.github.io/score-models/notebooks/01-score-function.html Supplemental Reading: Christopher Bishop (2023), *Deep Learning* Sec 20.3.1-20.3.3 *Lightning chat*: Sam & Dillon
Tues	Apr 7	Diffusion with Score Matching	Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33, 6840-6851. Supplemental topic: Diffusion and variational inference. Kingma, D., & Gao, R. (2023). Understanding diffusion objectives as the elbo with simple data augmentation. *Advances in Neural Information Processing Systems*, 36, 65484-65516 *Lightning chat*: Jacob & Mike

Thurs	Apr 9	Diffusion and SDEs	Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., & Poole, B. (2021). Score-based generative modeling through stochastic differential equations. In *Proceedings of the International Conference on Learning Representations (ICLR).* Supplemental Reading: Christopher Bishop (2023), *Deep Learning* Sec 20.3.4 *Lightning chat*: Owen, Ismoiljon

MODULE 2C: Diffusion with Transformers

Tues	Apr 14	Intro to Transformers	Christopher Bishop (2023), *Deep Learning* (Section 12.1: Attention). Secs. 12.2-12.4 are optional. Supplemental: Sasha Rush's *LLMs in 5 formulas* *Lightning chat*: Austin, Logan, Tatiana
Thurs	Apr 16	Attention Exercises	https://classic.d2l.ai/chapter_attention-mechanisms/index.html
Tues	Apr 21	Diffusion with transformers	Peebles, W., & Xie, S. (2023). Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 4195-4205). Supplemental: *An image is worth 16x16 words: Transformers for image recognition at scale.* Supplemental: Bishop *Deep Learning*, Ch 10: Convolutional Networks *Lightning chat: Diego, Aubrey*

MODULE 2D: Spatiotemporal Diffusion

Thurs	Apr 23	Conditional score-based diffusion	Tashiro, Y., Song, J., Song, Y., & Ermon, S. (2021). Csdi: Conditional score-based diffusion models for probabilistic time series imputation. *Advances in neural information processing systems*, 34, 24804-24816. *Lightning chat: Josh S, Josh C.* *Guest Lightning chat: Jack Ruder (Ph.D. Student)*
Tues	Apr 28	Probabilistic weather forecasting	Price, I., Sanchez-Gonzalez, A., Alet, F., Andersson, T. R., El-Kadi, A., Masters, D.... & Willson, M. (2025). Probabilistic weather forecasting with machine learning. *Nature*, 637(8044), 84-90. *Lightning chat: Lina, Laura*

MODULE 2E: Projects

Thur	Apr 30	Final project workshop	

Thurs	May 7	Project Submission Deadline

Note: The class schedule is tentative; it is subject to change as the course progresses.

References

[1] Louis Deslauriers, Ellen Schelew, and Carl Wieman. Improved learning in a large-enrollment physics class. *science*, 332(6031):862–864, 2011.

[2] Scott Freeman, Sarah L Eddy, Miles McDonough, Michelle K Smith, Nnadozie Okoroafor, Hannah Jordt, and Mary Pat Wenderoth. Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the national academy of sciences*, 111(23):8410–8415, 2014.

Final Project

During the period of April 29 to May 7 (9 days), you will be working in groups of 2 students on a project related to diffusion. Think of this project as an extended version of the "simple diffusion coding" project you did on March 31. The goal is to balance theoretical depth with a working implementation on some interesting dataset. Given the short timeframe and the "end-of-class" context, you should focus on feasibility (try to pick a relatively small dataset, so that you don't have to train a massive model from scratch).

I recommend progressing through the project as follows:

1. Project Scoping and Setup (Days 1–3)

- Define the Task: Choose a specific application such as image generation on a small dataset (e.g., MNIST, CIFAR-10), image inpainting, or unconditional generation for a niche domain like medical icons or textures.
- Read necessary material: If relevant, read through any background material that we didn't cover in class (e.g. U-nets for image modeling, DiffWave for audio modeling, etc.)
- Select a Framework: Use established libraries like PyTorch or Hugging Face Diffusers to avoid building low-level functions from scratch.
- Acquire Data: Ensure the dataset is small enough to train within 2–3 days on available hardware.

2. Implementation and Training (Days 4–6)

This phase involves the core technical work of building and refining the model.

- Build the Pipeline: Implement the forward diffusion process (adding noise) and the reverse process (denoising with some architecture, e.g. a U-Net or Transformer).
- Establish a Training Loop:
  - Sample a batch of observations and a uniform time step t.
  - Add Gaussian noise to the observations based on t.
  - Train the model to predict the added noise using Mean Squared Error (MSE) loss.
- Iterative Testing: Start with a tiny subset of data to ensure the loss decreases before committing to a full-scale training run.

3. Evaluation and Refinement (Days 7–8)

Once the model is training, focus on assessing its performance.

- Qualitative Analysis: Generate samples at various stages of training. How do they look? Document "hallucinations" or common failure modes.
- Quantitative Metrics: Use simple measures like training loss or MSE of the noise prediction to show convergence. To assess the quality of generated samples, you could try reporting standard metrics like FID (Fréchet Inception Distance).
- Hyperparameter Tuning: If time permits, experiment with different approaches (e.g. different model architectures or noise schedules, like linear vs. cosine).

4. Final Documentation (Day 9)

The final day should be dedicated to reporting results and ensuring the work is reproducible. Please submit a report (ideally no longer than 5 pages) that includes the following sections:

- Introduction. Clearly state the project goals and provide brief context for the problem being addressed.
- Methodology. Describe the model design, loss functions, and any data preprocessing steps (e.g., normalization, resizing). Include a link to your codebase to support reproducibility. Code should be well-organized and appropriately commented.
- Results. Present your findings, preferably with visualizations (e.g., plots or generated samples) to illustrate model performance.
- Conclusion. Summarize the key insights gained from the project. Discuss limitations and identify open questions or directions for future work.

Final Project Grading Rubric

Category	Points	Description
Introduction	10	Clear statement of project goals and motivation. Provides appropriate context.
Methodology	25	Model design, loss functions, and preprocessing steps are clearly described. Approach is reasonable and well-justified. Codebase is provided and organized.
Implementation & Effort	20	Evidence of a working implementation. Appropriate use of tools/frameworks. Demonstrates effort in building, debugging, and iterating on the model.
Results & Evaluation	25	Results are clearly presented (e.g., plots, samples). Includes some evaluation of model performance (qualitative and/or quantitative).
Conclusion & Insight	10	Clear summary of findings. Thoughtful discussion of limitations and possible improvements.
Clarity & Reproducibility	10	Paper is well-written and organized. Code is accessible, readable, and sufficiently documented to reproduce results.
Total	100	